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Database issue

OXFORD

RegulonDB v12.0: a comprehensive resource of transcriptional regulation in *E. coli* K-12

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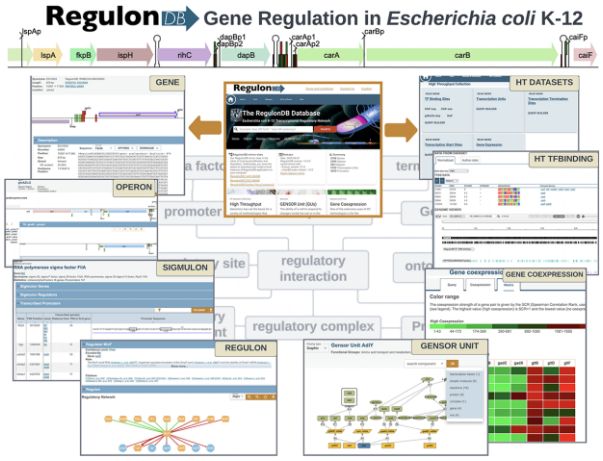
Abstract

RegulonDB is a database that contains the most comprehensive corpus of knowledge of the regulation of transcription initiation of *Escherichia coli* K-12, including data from both classical molecular biology and high-throughput methodologies. Here, we describe biological advances since our last NAR paper of 2019. We explain the changes to satisfy FAIR requirements. We also present a full reconstruction of the RegulonDB computational infrastructure, which has significantly improved data storage, retrieval and accessibility and thus supports a more intuitive and user-friendly experience. The integration of graphical tools provides clear visual representations of genetic regulation data, facilitating data interpretation and knowledge integration. RegulonDB version 12.0 can be accessed at https://regulondb.ccg.unam.mx.

Graphical abstract



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Introduction

Escherichia coli K-12 is one of the most widely used model organisms in research. Therefore, it is no surprise that the corpus of knowledge of regulatory mechanisms of gene regulation in this organism is among the best-characterized experimentally, defining the known fraction of the transcriptional regulatory network of *E. coli* K-12. Furthermore, knowledge of the regulatory network can be found in several databases, such as PRODORIC (1), which is focused on transcription factor (TF) binding motifs for different bacteria, RegPrecise (2), with microbial regulons reconstructed by comparative genomics, BioCyc (3), which includes a large number of a microbial genomes, and CoryRegNet (4) focused on corynebacterial gene networks, among several others. Unfortunately, it is not uncommon to find resources that do not keep updating their information. Other databases are dedicated to specific genomic methodologies such as proChIPdb (5) by the Palsson group and TEC with gSELEX data by the Ishihama group (6). The two most up-to-date databases with manually curated knowledge from the literature of *E. coli* K-12 are RegulonDB and EcoCyc. The knowledge of transcriptional regulation of these two databases comes from our curation team at UNAM.

The last NAR RegulonDB paper, on version 10.5, was published in 2019 (7), and in 2020 some of us formed a group of experts and published redefinitions of the fundamental concepts of the regulation of transcription initiation (8). As a consequence, we implemented several modifications in the database in order to align our concepts with these new definitions. Furthermore, as novel genomic technologies are increasingly expanding our knowledge, we have made significant efforts to gather all relevant knowledge coming from those methodologies. This specific progress was the basis for RegulonDB version 11.0 (9); however, that version was not reported in a NAR updated publication. Our efforts to normalize, gather metadata from, and make uniform these datasets contributed significantly to the comparability and usefulness of results coming from different laboratories using a variety of methodologies, as described in detail in (9). Thus, we describe the complete reconstruction and revision of the RegulonDB underlying data model and Web interfaces, and discuss the concomitant benefits. We have also complied with the findability, accessibility, interoperability, and reusability of data and software FAIR (10) requirements.

Materials and methods

RegulonDB software rebuild

RegulonDB is a database that provides information about the genetic regulation of *E. coli* K-12. It can be accessed at https://regulondb.ccg.unam.mx. The software architecture is based on the Model-View-Controller (MVC) pattern (11). The MVC pattern separates the code into three distinct layers with defined tasks, namely, Models, Views, and Controllers, where the database constitutes our model, the Web services encompass our controllers, and the components developed in React serve as the view layer.

Data processing. The initial phase of the release process is data processing. Multiple programs have been developed to handle data extraction, transformation, validation, and loading into RegulonDB. All programs were developed using Python programming language v3.11 (12) and Snakemake v7.14.0 as the workflow management system (13). The PythonCyc library was utilized to extract data from EcoCyc (14). EcoCyc is a bioinformatics database that describes the genome and the biochemical machinery of *E. coli* K-12 MG1655 and the knowledge of transcriptional regulation and operon organization stemming from our curation work.

Database and datamarts. RegulonDB was implemented using MongoDB v6.0 (https://www.mongodb.com/), a NoSQL, document-oriented database manager that offers the advantage of a flexible data schema. This flexibility allows us to adapt to the heterogeneity of information common to biological knowledge. In our MongoDB implementation, each concept from our model is represented as a separate collection. These collections collectively form the normalized database, where we carefully address information redundancy to maintain data integrity and efficiency. However, since the interfaces are designed to display information from multiple collections (concepts), we have incorporated datamarts.

A datamart is a specialized data storage system that focuses on specific subject areas, enabling efficient analysis of subject-specific information (15). It provides integrated and summarized data to support targeted queries. In RegulonDB, we have created five distinct datamarts, each specifically tailored to serve as the underlying data source for various search result interfaces, including gene, operon, regulon, sigmulon, and genetic sensory-response units (GENSOR units).

An API was created for querying, joining and transforming data to populate the datamarts. Each datamart is designed to cater to a specific query object within the Web interface.

Web services. Access to data was implemented through Web services that use GraphQL v15.5.0 (16), Apollo Server Express v2.21.0 and Node v16.13.0. GraphQL is a query language for APIs designed for data structures in the form of trees or hierarchies, which facilitate how information is queried. Currently, the request makes it possible to specify which data are desired and from which level; the output format is JSON (17).

Web interface. A component-based Web interface was developed using ReactJS v17.0.2 (https://es.reactjs.org/), an open-source JavaScript library created by Facebook. ReactJS offers development benefits such as flexibility, speed, and code organization, making it one of the best tools for this purpose. In addition, third-party libraries such as material-ui (https://mui.com/material-ui/), svg.js (https://svgjs.dev/docs/3.0/), react-table (https://tanstack.com/table/v8) and cytoscape.js (18) (https://js.cytoscape.org/) are used.

All software developed by RegulonDB complies with the FAIR software guidelines (10,19). The entire code is registered on FigShare. The software development team has followed the software development recommendations of the Moprosoft level 2 process model (20).

Update and expansion of evidence codes and confidence levels

We updated the evidence codes used in RegulonDB with the aims of both increasing their precision to help distinguish the different methods, and to expand them to capture the variety of high-throughput (HT) methods used. We used the ECO ontology (21) as well as the evidence in EcoCyc (22). When possible, we used the same codes as those in ECO, although several of our codes are not present in that ontology. This work was done by our team, with input from interactions with the

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EcoCyc team. Our updated evidence codes are now present in both RegulonDB and in EcoCyc.

A unique feature in RegulonDB is the existence of confidence levels, which have been available since 2013 (23). Essentially, we classify the set of evidence for a given object in three major categories: weak, strong, or confirmed. Certainly, some methods provide physical evidence for the existence of an object (i.e. Northern blotting for a transcription unit [TU], or DNAse footprinting for a binding site), whereas others are less reliable (i.e. computational predictions, or binding of cellular extracts). Furthermore, it is rather frequent that a given object has been identified by more than one method. In our initial paper (2013), we explained the criteria to identify independent methods, so that whenever an object had evidence from independent methods, this information contributed to increase the confidence level of a given object. This algebra, so to speak, also has been also updated, given the increase in evidence codes. For more details, see (24).

Results

We present our progress in two sections. First, we describe novelties of biological additions to the database. Subsequently, we describe the software reconstruction supporting RegulonDB.

Updated content

Selected curation additions

Since the latest version of RegulonDB, published in NAR in 2019 (7), several types of data have been added by means of our curation of low-throughput data. In the following, we summarize a selection of specific additions to the database. We modified several promoters given the new definition in the Mejía-Almonte *et al.* review (8). If a promoter was recognized by two different sigma factors and initiated transcription at the same transcription start site (TSS), it was separated into two promoters each with its own sigma factor and with the same TSS coordinate. As a result, we created many new promoters and thereby new TUs. Also, a relevant distinction was made between transcription factor binding sites (TFBSs) and the subset of those that have evidence not only of binding but also of a regulatory effect, now called transcription factor regulatory sites (TFRSs).

It is not so common to add new genes in the highly studied *E. coli* genome, for which there are currently >4500 known genes. During this period, genes encoding a pair of new small proteins XtpA (25) and SpfP and several small RNAs, such as GlnZ (26), StfZ (27) and AmeF (28), among others, have been added, including the regulatory interactions (RIs) of the new small RNAs.

CRP is a global TF that regulates a large number of genes, and it has been extensively analyzed. Recently, a CRP complex with the polypeptide Sxy was identified. The database now includes the CRP-Sxy complex, which is responsible for regulating certain genes associated with DNA uptake (natural competence). As described by Søndberg *et al.* (29), this CRP-Sxy complex recognizes a DNA-binding site with an asymmetric arrangement: it comprises two separate half-sites, with one resembling the typical CRP-cAMP DNA-binding site, while the other half-site shows lower conservation. Other new conformations for other TFs have been also added to the database, such as dimer formations for FrlR (30), DgoR (31), NanR (32), ChbR (33), SlyA (34) and RcdA along with newly identified inactive configurations, like FrlR-fructoselysine-6-phosphate (30), MprA-salicylate (35), Cbl-thiosulfate (36), MurR-*N*-acetyl-D-glucosamine-6-phosphate (37) and AraC-D-fucose (38,39). In addition, the acetylated conformation of RcsB has been annotated (40).

We added the homodimer form as well as the inactive ligand-bound forms of the TFs RcdA, RcdA-Tris and RcdA–trimethylamine N-oxide (41). Four new inactive conformations of the TF AcrR were included, such as AcrR-3,6-diaminoacridine (proflavin) and AcrR-rhodamine 6G (RDG)(42,43). On the other hand, Galego et al. demonstrated that the functional activity of the TF BolA relies on its phosphorylated state (44). All of the previous RIs of BolA are now associated exclusively with the active BolA-phosphate form.

Some proteins have recently been found to have the function of a TF, such as the sigma factor RpoN, which was described by Shimada et al. as a transcriptional repressor of Sigma70 promoters in the absence of enhancers (45). Turnbull and Gerdes identified the HicB protein, known as the antitoxin of the HicA-HicB toxin-antitoxin system, as a TF that autoregulates the transcription of the hicAB operon (46). On the other hand, proteins that had been predicted as TFs were confirmed as proteins with that function upon the discovery of their first regulatory RIs by HT-binding and expression analyses; a few instances of these novel TFs include YfeC, YidZ, YciT, and YgbI (47); AaeR (48); PunR (49); YieP, YeiE, and YafC (50); YiaU (51); YgfI (52) and FimZ (53). Finally, the DNA-binding consensus sequences of 125 TFs and five sigma factors have been annotated in the database (54).

Duarte-Velázquez *et al.* (55) described 58 potential TFs, including details like their protein family classification, DNA-binding domains, potential regulated processes, and the mechanisms governing their regulation (56). We have included these descriptions in the summary for each putative TF.

Furthermore, we performed a detailed curation of the whole repertoire of known allosteric effectors, currently for 90 TFs, most of which respond to only one allosteric metabolite, with few known cases sensing two or even more different metabolites (see Figure 3 and supplementary material in (57)). As a consequence, all effectors and TF-effector conformations now have the full name of the effector (i.e. GalR-D-galactose), both in RegulonDB and in EcoCyc.

During this period, some modifications were made to the evidence codes and evidence ontology, leading to the creation of 14 new and more specific evidence codes and the reclassification of six existing ones. These evidence codes pertain to distinct experimental approaches, encompassing both traditional and HT methods, and are currently employed in the curation process. See Evidence Classification in the Help menu (24).

The total number of current objects and the total increase of objects compared to version 10.5 are shown in Table 1. The historical increase can be found in RegulonDB in the *Release* option in the menu releases.

HT dataset collection updates

One of the major emphases in our recent work has been the addition and nearly up-to-date completion of all published HT genomic datasets containing TSSs (16 datasets), transcription termination sites (5 datasets), TUs (5 datasets), RNA-sequencing (RNA-seq) (1864 datasets) expression profiles and HT binding methods supporting TFBSs (29 datasets

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Table 1. Total objects in version 12.0 in the first column, and total increases since version 10.5

Object	Total objects	Total increases in no. of objects
Promoters	4050	490
Regulatory Interactions	6110	2644
Regulatory sites	4361	1341
Transcription units	3705	152
Transcription factor	238	23
Terminators	371	11

In addition to these objects, there has been an increase in the following: GUs (185), small RNAs (2) and genes (60). The increase in promoters is determined after the elimination of a TSS HT subset (described in the text).

from chromatin immunoprecipitation assays combined with sequencing [ChIP-seq], 94 from combined with lambda exonuclease digestion and next-generation sequencing [ChIP-exo], 164 from genomic SELEX screening [gSELEX] and 107 from DNA affinity purification sequencing [DAP-seq]). Because each experiment generates a collection of data, we implemented a separate section out of the standard computational container (previously a relational database and now a document-based Mongo database) where users can access all of them. As a consequence, we transferred the sets of TSSs identified by the Morett group from the core of RegulonDB to the HT dataset collection. This explains the considerable decrease of promoters after version 11.1 that can be observed in the summary history of total objects through time (see the 'Releases and Downloads' menu). Most of the HT datasets come from manual curation of the corresponding publications, whereas the raw data of these collections are found at repositories such as GEO and ArrayExpress or from dedicated databases such as the Palsson´s proChIPdb (5) with their ChIP-exo experiments (https://prochipdb.org/dataset_page.html?organism=e_coli) and Ishihama´s TEC database for gSELEX (https://prochipdb.org/dataset_page.html?organism=e_coli). It is important to emphasize that we did not simply import such data, but in most cases this effort involved a considerable amount of manual curation, including direct interactions with some authors, as well as the processing, normalizing the data, or even running our own pipeline (58) to enhance their comparability. Such is the case of the uniformly filtered and processed collection of ChIP-seq data as well as the large RNA-seq profiles collection (for details see (9)). We implemented bidirectional links between these datasets and the corresponding TF gene and regulon pages in RegulonDB.

The HT-TFBS dataset collection has recently been updated, so that now all peaks and sites align with the U00096.3 version of the genome, enabling comparisons among the datasets as well as with the regulatory interaction (RI) set in RegulonDB at the coordinate level. These comparisons enabled us to add a column in every HT-binding dataset indicating those peaks or sites that match with a known regulatory site in RegulonDB. Additionally, we have incorporated the HT-binding evidence into the corresponding RIs; with our rules identifying independent methods, we recalculated their confidence level (for details, see (24)).

We performed a comparative analysis of the datasets processed by our pipeline versus the same datasets processed by the authors from the ChIP-seq collection; this comparison was at the peak coordinates level and, as detailed in the supplementary material, we found that our pipeline recovered a good proportion of peaks identified by the authors (average of 86%), and the pipelines by authors as well as our recovered close to 50% of the RegulonDB TFRS collection containing at least one classical evidence (see the supplementary material).

The HT collection was recently enriched with a new subcollection of genome-wide binding sites from the RNAP (RpoB) (6 datasets) and sigma factors (22 datasets). In the case of data from ChIP-seq, ChIP-exo and ChIP-CHIP they were curated from publications and in the case of gSELEX datasets from the TEC database. They can be found within the HT-binding collections of the corresponding methodologies.

New version of position-specific scoring matrices (PSSMs)

Given the considerable increase in the number of TFRs, we have regenerated the whole collection of PSSMs using as input the annotated TFRSs from the current version 12.0 of RegulonDB. We added 3 bp to each side of the reported sites since the first step is the multiple alignment of the collection of sites for each TF. We used command-line MEME (59) (https://meme-suite.org/meme/tools/meme) version 5.4.0. We obtained the median width for the TFRSs of each TF and allowed MEME to search motifs with a width of the median +/- 3 base pairs. MEME was run in mode 'One Occurrence Per Sequence' and the reference Markov background model file was included for normalization. We were able to generate a matrix for 109 TFs out of the 238 total current TFs in our collection. Not all TFs have enough sites to support a matrix model given the parameters used in MEME. The matrices can be accessed in the portal within the 'Releases and Downloads' menu. When comparing the new and previous versions we did not see major differences, although we did not make a precise comparison. We also did not see notorious differences between the matrices generated using either the TFRSs with strong or confirmed evidence or using all TFRSs. The previous matrices can be accessed in the Releases & DownLoads menu.

Lisen&curate: a curation platform to facilitate gathering textual evidence for curation and annotation of phrases from scientific publications displayed in RegulonDB.

Lisen&Curate is a curation platform we built to facilitate linking sentences or part of sentences, which we call 'sources' present in the scientific publications that we curate, with their corresponding curated objects, facilitating a direct link for users to more quickly access the original sentences supporting the knowledge encoded in RegulonDB (60). These sources are offered within RegulonDB. Currently, there are 23 sources linked to promoters, 157 to TFs, 179 to RIs,149 to TFRS, and 126 to TUs.

This platform includes an interface to annotate growth conditions and link them to the different objects of the database. The interface has integrated the Microbial Conditions Ontol-

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ogy (MCO) (56) so curators use its controlled vocabulary. The MCO format includes the pertinent features for reproducibility, that we gather from the literature, when provided by the authors, such as growth media, supplement, temperature, genetic background, growth rate, etc.) (see the MCO browser from Tools in the RegulonDB menu).

RegulonDB software rebuild

Data processing

RegulonDB collects data from multiple sources, including literature curation, which contributes to both RegulonDB and EcoCyc. We ensure a close coordination of data and updates with EcoCyc to maintain accurate and up-to-date information. The current version, 12.0, is synchronized with the EcoCyc 27.0 release. Another source of data comes from our internal curation using Lisen&Curate, where source sentences from articles supporting the annotated information, as well as growth conditions and other relevant data, are curated. A separate recent relevant source is that of our processed large collections of HT-generated datasets, as mentioned before. Finally, we also feed data from datasets provided by investigators, who supply the data in a specific format that we require for their integration.

RegulonDB periodically releases multiple data updates throughout the year, which can be accessed from the Releases & DownLoads option in the main menu. In recent versions, we have implemented Semantic Versioning as a standardized approach. These data are versioned using a three-component number system, represented as X.Y.Z. The major version (X} is used for significant updates and the addition of important biological content, typically every 2 years. The minor version (Y) is employed for quarterly releases or updates to data for all elements of a biological object. The patch version (Z) is applied for addressing minor data errors. This approach ensures improved tracking and management of data updates in RegulonDB.

An important aspect in achieving discoverability of information (61) is the assignment of a unique identifier. In RegulonDB, we have allocated unique identifiers to each biological element stored as a document (record) by using the pattern RDB[A-Z]{8}[0–9]{5}, where RDB represents the RegulonDB acronym, the following eight letters correspond to the organism, biological object, and type of dataset (for example, ECOLIPMC corresponds to an *E. coli* promoter from classical data), and finally the subsequent 5-digit number represents a consecutive identifier. Furthermore, for achieving interoperability, we integrate the identifiers from the data sources with which we collaborate.

The release process starts with the data processing phase, during which data from all of our contributing sources are processed. Several Python programs and workflows in Snakemake have been developed for extracting, transforming, validating, integrating, and loading the information into RegulonDB.

Database and datamarts

To encode in our database a large range of different degrees of knowledge of the different regulatory elements, either highly or poorly characterized, we migrated from a relational database to a flexible model that can adapt to data gaps and constant changes in the data model. Specifically, MongoDB, a document-oriented NoSQL database, was chosen. In this infrastructure, data are stored in documents instead of records. These documents are stored in BSON, a binary JSON representation.

Unlike relational databases, MongoDB offers greater flexibility and performance in handling data. Its ability to store and query unstructured or semistructured data allows for efficient handling of information gaps. Additionally, MongoDB easily adapts to the changing dynamics of the data model, providing a robust and scalable solution for the storage and manipulation of massive data. In order to integrate information from various data sources and to implement constraints to safeguard data integrity, a normalized form of collections was implemented in MongoDB, maintaining references among them.

Figure 1A describes the conceptual model of the database in RegulonDB, where each class represents a biological object and these classes are referred to as collections in MongoDB. Each collection is related through foreign keys. On the other hand, Figure 1B represents the RegulonDB datamarts, which are integrated collections that combine multiple biological concepts and are designed to meet the requirements of the main search result pages on the Website. For instance, the gene datamart integrates and displays information about the gene itself and its product in the gene Web page. Similarly, the operon datamart incorporates details regarding the operon, its TUs, promoters, terminators, and operon regulation. In the case of the regulon datamart, information about the regulator, its targets, and RIs is displayed (see Figure 1B).

Web services

In order to access the data in RegulonDB, Web services (WS) were implemented. A Web service uses a set of protocols and standards to exchange data interoperably between applications. Interoperability is achieved through the adoption of open standards.

For each datamart in RegulonDB, there are two main Web services: getBy and getAll. For example, for 'gene' we have 'getGenesBy' to search for genes with filtering options and 'getAllGenes' which allows extraction of all the documents from a collection. The getBy Web services have two modes of search: simple and complex. Both modes allow for logical operators and result pagination. In advanced searches, it is possible to specify fields and filters.

Currently, RegulonDB has implemented 46 Web services that are associated with its datamarts, tools and HT data collections. These Web services can be accessed through the Search menu under 'Search by WS', and users can utilize the playground tool to conduct queries. The playground consists of three crucial sections: the Query area, the Results area and the Web services catalog, as shown in Figure 2.

Web interfaces

Over the years, we have learned that one of the greatest challenges in maintaining public databases is the application maintenance. RegulonDB, developed with over 20 years of software development experience, eventually became an infrastructure built on various technologies, some of which became obsolete.

To modernize the infrastructure and applications of RegulonDB, a comprehensive rebuild effort was undertaken using a component-based development approach. This approach emphasizes the reuse of components to enhance maintenance and expedite the development of new applications. Figure 3 shows

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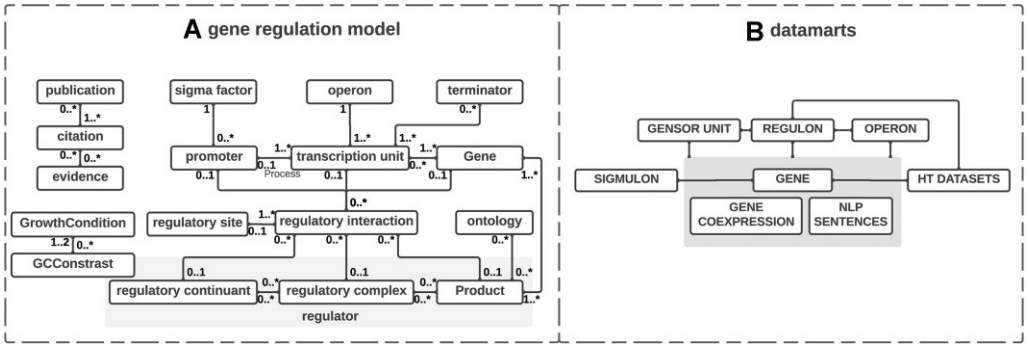

Figure 1. Class diagram of genetic regulation and datamarts in RegulonDB. (**A**) Class model of gene regulation. In this diagram, each box represents an entity or biological concept, and the lines between them indicate cardinality relationships. The notation '1..*' represents a relationship of one or more elements, while '0..*' indicates a relationship of zero or more elements, that is, optional elements. (**B**) Class diagram of the datamarts in RegulonDB. These datamarts integrate one or more biological entities from the class model presented in panel a. This graph also describes the RegulonDB Web page navigation.

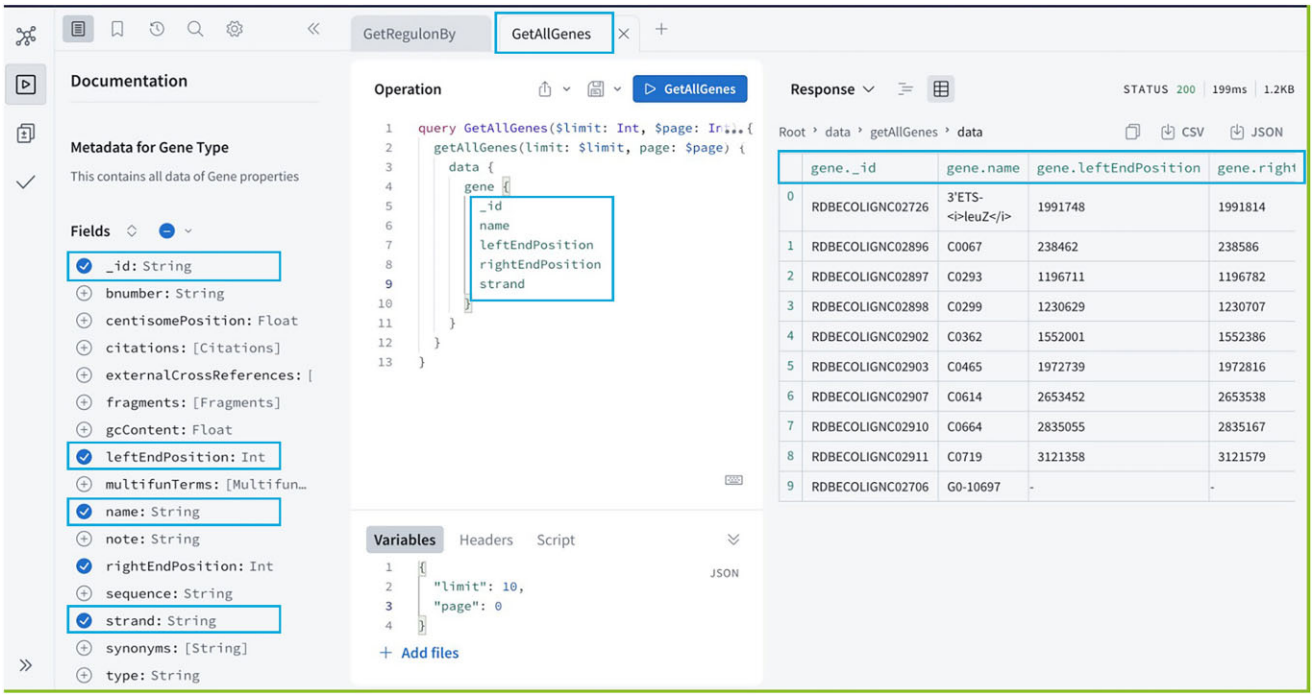

Figure 2. Illustration of the apollo sandbox interface to perform searches in RegulonDB using Web services. At point 1, in the query section, users can select the attributes they wish to visualize. At point 2, the query results are displayed based on the applied filters. At the final point, marked as 3, all available Web services are listed.

several essential components used in the pages, including the query box, menu, main object data, graphic display, navigation menu section, related tools menu and the data body section.

The graphical representation of gene regulation has been a central focus of our software development. The Drawing Traces tool (DTT) generates visual representations of genomic regions with annotated genetic elements. The DTT has been reengineered using components in ReactJS, leading to the creation of a comprehensive library where each component caters to the graphical representation of specific elements, including genes, operons, promoters, binding sites, terminators, and more. The functionality of this tool, along with the utilization of the graphical library, can be observed and explored on the gene and operon pages.

The five main Web pages (gene, operon, regulon, sigmulon, and GENSOR units) were redesigned to follow the same structure for a smoother browsing experience. As shown in Figure 3, they provide two options for searching, one using the box on the top right (see component 1 in the figure) when the user knows what object they are looking for, and the other on the left (see component 2 in the figure) that lists all objects to choose from.

RegulonDB FAIRification

We have implemented the FAIR principles to enhance the findability, accessibility, interoperability, and reusability of data and software in RegulonDB. The current version of RegulonDB´s data has been registered in Zenodo with their citation and a permanent link to this version, thus guaranteeing their

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Figure 3. illustrates the main Web components utilized in creating the five key Web pages mentioned: (1) the query box; (2) the catalog of elements where users can browse and search for specific items; (3) the object identification properties component; (4) the graphical representation component of the queried object; (5) the navigation menu for moving between sections of the page; (6) the related tools component and (7) the component describing object specific details.

findability and accessibility. We used FAIRSharing to register the complete description of RegulonDB as a database, contributing to enhance its visibility and easy understanding by the scientific community. FAIRSharing is a platform of educational and informative resource for databases and their use. We improved the accessibility of data generating alternative options to reach them. In addition to the portal interfaces, we implemented Web services through GraphQL, which generates answers in JSON format. This facilitates users to access data in RegulonDB by directly using it in different programming languages such as Python, R, Java and Ruby, among others.

To enhance the interoperability of data we adapted standards and well-known ontologies useful for knowledge of transcriptional regulation. We are using the MCO as mentioned, to describe growth conditions and genetic background used in the experiments, the Gene Ontology for the functional annotation of genes, and the ECO ontology as a reference in our updated evidence codes. Additionally, we are building an ontology of the major concepts of the regulation of transcription initiation formalizing the updated definitions already mentioned. The use of ontologies enhances the interoperability of data, because the same concepts may be used by other bioinformatics resources.

We made sure to offer the necessary information concerning the licenses for data usage. These can be found in the RegulonDB portal, and in Zenodo. We paid attention also to follow standards in the metadata descriptions and in the most frequently requested downloadable files.

Concerning software, we applied similar principles of accessibility and reusability. All our scripts and computational tools are publicly available in GitHub facilitating their access and providing an easy way to download them. We use Git for version control and we offer a usage license for each project. We registered in FigShare an integrated version of our application which contains the database, Web services and the graphical interfaces.

We are aware that there is still room for improvements, particularly the refactorization of code to generate reusable libraries to enable maintenance and interoperability across other tools, platforms and resources.

Briefly, in the rebuild of our interface and new data version we have implemented the four FAIR principles for data and software using registers, standards, ontologies and options for adequate licensing of RegulonDB. We are aware and committed to maintain a constant work as standards and requirements continue to evolve within the scientific community.

Docker container

To facilitate the distribution and use of RegulonDB, the Docker tool has been used to package the datamarts, the Web service API and the Web application, along with all of its dependencies. The resulting Docker container can be downloaded from the 'API & Software' menu of RegulonDB and installed on any computer. Once installed, the Web application can be accessed from a browser, simplifying its implementation and access without the need for complicated configurations.

Discussion

As is usual in this annual NAR database issue, we have focused on presenting the updates in biological content, design and

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bioinformatics infrastructure since our last NAR publication in 2019.

We describe in detail a selected sample of novel biological content gathered from our constant manual curation. Furthermore, as mentioned, we made changes to the organization of knowledge to fit with the redefined concepts as discussed and supported in a recent publication (8). We updated the PSSM collection including now matrices for 109 TFs. As mentioned, one of the major improvements in this period has been the addition of published HT genomic datasets. They can be either directly accessed going in the main page to the HT collections, or through links in the Web pages of the relevant objects. We report our update of evidence codes and corresponding confidence levels. We know RegulonDB knowledge is frequently used to compare with HT novel binding technologies. With our evidence update users will be able to filter, for instance RIs excluding specific methods, to prevent circularity in their benchmarking (see (24)).

Unfortunately, the RNA-seq collection has been hard to integrate with additional knowledge in the database. Certainly, despite all our efforts using natural language processing strategies, we are far from completing the metadata that describe the growth conditions for each dataset; these are essential to link RNA-seq profiles with, for instance, specific TFs. As detailed in (9), we know that a large fraction of metadata is missing in GEO SOFT files and even in scientific publications as noted since the MCO implementation (56). We currently do not have any metadata for 416 SRRs (SRA accession IDs) of 1289 we gathered. Thus, these 416 SRRs can only be searched based on their SRR ID in the RegulonDB HT collection.

As far as we know, RegulonDB is the only microbial database that strives to maintain up to date in such a comprehensive way experimental data coming both from classic molecular biology methods as well as from the more recent HT-related wave of methods.

The reconstruction of RegulonDB's computational infrastructure has allowed significant improvements, facilitating access to data. We have implemented changes to satisfy FAIR principles in RegulonDB, and we continue to work on improving interoperability and reusability. The transition from a relational model to a document model with MongoDB will facilitate the storage and retrieval of complex data related to gene regulation, making future database scalability easier. In fact, NoSQL systems have been used in various bioinformatics resources (62–64).

Furthermore, the use of graphQL queries has improved efficiency and flexibility in data access. The design of the RegulonDB interface has been updated using React to enhance user experience and enable more intuitive navigation. Some graphical tools have been developed and added, leveraging a component-based development approach. These graphical tools provide a clearer and more understandable visualization of genetic regulation data, making knowledge integration easier. Despite our current progress, there are still some modules to be reconstructed, and we continue to work on improving the interfaces. We are also exploring new technologies related to artificial intelligence to facilitate the curation process.

Overall, the improved computational improvements, the enhanced quality of knowledge representation, the up-to-date knowledge of decades of experimental work by molecular biology methods gathered through decades of manual curation, plus the comprehensive collection of HT datasets of the main components of regulation of transcription initiation make RegulonDB a highly valuable and unique bioinformatics platform to enhance our understanding of microbial gene regulation.

Data availability

All RegulonDB programs are available as open source under Apache License 2.0 license. Users can access them in the RegulonDB software project repository on GitHub at the following link: https://github.com/regulondbunam/regulondb-releases/. This license allows users to use, modify, and share the code openly as long as it is properly attributed to RegulonDB. The release data version 12.0 of RegulonDB have been registered in Zenodo with DOI 10.5281/zenodo.8346680 and the entire code is in FigShare with DOI 10.6084/m9.figshare.24129402.

Supplementary data

Supplementary Data are available at NAR Online.

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Conflict of interest statement

None declared.

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